Chapter 1: Basic Finance

Abaynew Z.

July 29, 2025

Introduction

Finance is defined as the art and science of managing money. It includes both financial services and financial instruments. Finance also refers to the provision of money at the time when it is needed, ensuring that funds are available for investments, operations, or other activities. The concept of finance broadly covers key elements such as **capital**, **funds**, **money**, and **amount**, all of which are essential for economic and business decision-making.

Finance is one of the important and integral parts of business concerns. It plays a major role in every aspect of business activities, from planning and operations to decision-making and long-term growth.

Finance can broadly be classified into **two** major categories: **Private Finance** and **Public Finance**. These two categories distinguish the financial activities carried out by individuals and private entities from those managed by government bodies and public institutions.

Private Finance

Private Finance refers to the financial activities conducted by individuals, firms, businesses, and corporations to manage their resources and meet their financial goals.

It includes

- **individuals**, such as saving money in a bank or investing in stocks for retirement.
  Example: A person saving money in a bank or investing in stocks for retirement.
- **firms**, for instance, when a small manufacturing firm decides whether to take a loan to buy new machinery.
  Example: A small manufacturing firm taking a loan to buy equipment.
- **businesses**, private finance includes managing working capital and ensuring daily cash flow is sustained.
  Example: A retail business managing its cash flow.
- **corporate financial activities** required to meet broader business goals, such as issuing bonds or shares to raise capital for expansion.
  Example: A corporation issuing bonds for growth.

Public Finance

Public Finance concerns the revenue collection and expenditure activities of the government. It includes financial matters at different levels of government.

It includes

- **central government** level, public finance involves preparing the national budget, collecting taxes, and allocating funds for defense and infrastructure.
  Example: Preparing the national budget and funding public projects.

- **state government** level, it includes managing education budgets, state road construction, and state-level taxation and subsidies.
  Example: Managing state education and infrastructure budgets.

- **semi-government** financial matters pertain to the financial operations of government-owned corporations such as Ethiopian Airlines or telecom companies, covering pricing policies and investment decisions.
  Example: Pricing and investment decisions of government-owned enterprises.

Financial Mathematics

The field within finance that draws on the use of quantitative methods is called mathematical finance. **Financial mathematics** involves the application of mathematical techniques to solve problems in finance and investment. It describes how **mathematics** and **mathematical modeling** are used to analyze and address various financial challenges, such as pricing financial instruments, managing risk, and optimizing portfolios. This field is often referred to as ***quantitative finance***, ***financial engineering***, or ***computational finance***, reflecting its interdisciplinary nature and reliance on mathematical and statistical tools.

1.1 Interest

Definition: Interest is the cost of borrowing money or the return on investment, expressed as a percentage of the principal over time.

Formulas:

- Simple Interest: $I = P \times r \times t$
- Compound Interest: $A = P(1 + r)^t$

where

- P = Principal
- r = interest rate per period
- t = time in years
- A = amount after interest

Example 1: Simple Interest

Calculate simple interest on \$5,000 at 6% annual rate for 3 years.

Solution:

$$I = 5000 \times 0.06 \times 3 = 900$$

Thus, the interest is \$900 after 3 years.

Example 2: Simple Interest

Calculate the simple interest on a loan of \$8,000 at an annual interest rate of 7% for 4 years.

Solution:

Simple Interest formula:

$$I = P \times r \times t$$

where

$$P = 8000, \quad r = 0.07, \quad t = 4$$

Thus,

$$I = 8000 \times 0.07 \times 4 = 2240$$

The simple interest is \$2,240.

Example 3: Simple Interest

If you invest \$5,500 at a simple interest rate of 5% per year, how much interest will you earn after 3 years?

Solution:

$$I = 5500 \times 0.05 \times 3 = 825$$

You will earn \$825 in interest after 3 years.

Example 4: Compound Interest

Find the compound amount for \$6,000 at 6% annual interest compounded annually for 3 years.

Solution:

Compound Interest formula:

$$A = P(1 + r)^t$$

where

$$P = 6000, \quad r = 0.06, \quad t = 3$$

Calculation:

$$A = 6000 \times (1 + 0.06)^3 = 6000 \times 1.191016 = 7146.10$$

The compound interest:

$$I = A - P = 7146.10 - 6000 = 1146.10$$

The compound interest is \$1,146.10, and the total amount is \$7,146.10.

Example 5: Compound Interest

You deposit \$10,000 in an account with an annual interest rate of 8%, compounded annually. What will be the amount after 5 years?

Solution:

$$A = 10000 \times (1 + 0.08)^5 = 10000 \times 1.469328 = 14693.28$$

Compound interest:

$$I = 14693.28 - 10000 = 4693.28$$

The compound interest earned is \$4,693.28, and the total amount is \$14,693.28.

Example 6: Compound Interest (Semi-Annual)

Calculate the compound amount on \$4,000 at 6% annual interest compounded semi-annually for 2 years.

Solution:

For semi-annual compounding:

$$r = \frac{0.06}{2} = 0.03, \quad n = 2 \times 2 = 4$$

Thus:

$$A = 4000 \times (1 + 0.03)^4$$

Calculation:

$$(1.03)^4 = 1.1255$$

$$A = 4000 \times 1.1255 = 4502$$

Compound interest:

$$I = 4502 - 4000 = 502$$

The compound interest is \$502, and the total amount is \$4,502.

Example 7: Comparing Simple vs. Compound Interest

You invest \$3,000 at 5% for 3 years. Find the interest using:

(a) Simple Interest

(b) Compound Interest (annually)

Solution:

(a) Simple Interest:

$$I = 3000 \times 0.05 \times 3 = 450$$

\$450 simple interest.

(b) Compound Interest:

$$A = 3000 \times (1 + 0.05)^3 = 3000 \times 1.157625 = 3472.88$$

$$I = 3472.88 - 3000 = 472.88$$

\$472.88 compound interest.

Observation: Compound interest (\$472.88) is slightly higher than simple interest (\$450) over 3 years.

Exercise

1. Find the simple interest on Birr 3,000 at 7% for 1 year.
   Ans. = Birr 210

2. Find the simple interest on Birr 10,000 at 12% for 6 months (0.5 year).
   Ans. = Birr 600

3. Find the rate if a loan of Birr 20,000 incurs interest of Birr 12,000 after 8 years.
   Ans. = 7.5%

4. Find the time required for Birr 50,000 to earn Birr 10,000 at 8% interest per annum.
   Ans. = 2.5 years

5. Deposit 4000 birr at 6% interest compounded quarterly for 5 years.
   Ans. = Birr 1387.42

6. Find present value needed to get 12,000 birr in 6 years at 9% compounded monthly.
   Ans. = Birr 7002.63

7. Time required for 5000 birr to grow to 8000 birr at 6% compounded monthly.
   Ans. = 7.84 years

1.2 Inflation

Definition: Inflation is the rate at which the general price level rises, reducing money's purchasing power.

Inflation Rate

The inflation rate measures the percentage increase in the price level over a period:

$$\text{Inflation Rate} = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\%$$

where P_t is the price index at time t and P_{t-1} is the price index in the previous period.

Example: If the Consumer Price Index (CPI) increases from 200 to 210,

$$\text{Inflation Rate} = \frac{210 - 200}{200} \times 100\% = 5\%.$$

Future Value with Inflation

To compute the future value adjusted for inflation, the general formula is:

$$FV = PV \times (1 + i)^n$$

where:

- FV = future value
- PV = present value
- i = inflation rate (decimal form)
- n = number of periods

Example 1

If inflation is 5% annually, find the price of a $200 item after 2 years.
Solution:

$$\text{Future Price} = 200 \times (1 + 0.05)^2 = 200 \times 1.1025 = 220.50$$

Thus, the price will be $220.50 after 2 years.

Example 2

If the inflation rate is 5% per year, what will be the price of an item currently costing $200 after 3 years?
Solution:

$$r = 0.05, \quad P_0 = 200, \quad t = 3$$

$$P_t = 200 \times (1 + 0.05)^3 = 200 \times 1.157625 = 231.53$$

The price after 3 years will be $231.53.

Example 3

A textbook currently costs $750. If the annual inflation rate is 6%, find its price after 4 years.
Solution:

$$r = 0.06, \quad P_0 = 750, \quad t = 4$$

$$P_t = 750 \times (1 + 0.06)^4 = 750 \times 1.262477 = 946.86$$

The price after 4 years will be $946.86.

Example 4

The price of a bag of cement is currently \$450. If the inflation rate is 4% per year, find its price after 5 years.

Solution:

$$r = 0.04, \quad P_0 = 450, \quad t = 5$$

$$P_t = 450 \times (1 + 0.04)^5 = 450 \times 1.2166529 = 547.49$$

The price after 5 years will be \$547.49.

Example 5

A bus ticket costs \$30 now. If inflation is 3% per year, what will it cost after 6 years?

Solution:

$$r = 0.03, \quad P_0 = 30, \quad t = 6$$

$$P_t = 30 \times (1 + 0.03)^6 = 30 \times 1.194052 = 35.82$$

The price after 6 years will be \$35.82.

Example 6

An apartment currently rents for \$2,000 per month. If the annual inflation rate is 8%, what will the rent be after 2 years?

Solution:

$$r = 0.08, \quad P_0 = 2000, \quad t = 2$$

$$P_t = 2000 \times (1 + 0.08)^2 = 2000 \times 1.1664 = 2332.80$$

The rent after 2 years will be \$2,332.80.

Example 7

The price of a computer is \$1,500 now. If the annual inflation rate is 7%, what will it be after 3 years?

Solution:

$$r = 0.07, \quad P_0 = 1500, \quad t = 3$$

$$P_t = 1500 \times (1 + 0.07)^3 = 1500 \times 1.225043 = 1837.57$$

The price after 3 years will be \$1,837.57.

Present Value with Inflation

To find the present value from the future value under inflation, the formula is:

$$PV = \frac{FV}{(1+i)^n}$$

where:

- PV = present value
- FV = future value
- i = inflation rate (decimal)
- n = number of periods

Example: If you expect to need 1157.63 after 3 years with 5% annual inflation, the present value required is:

$$PV = \frac{1157.63}{(1 + 0.05)^3} = \frac{1157.63}{1.157625} = 1000.$$

This means you need to invest 1000 today to match the future value required under 5% annual inflation over 3 years.

Causes of Inflation

Introduction

Inflation is a sustained increase in the general price level of goods and services in an economy over time. The causes of inflation are typically grouped into three major categories: **demand-side**(also called demand-pull), **supply-side** (also called cost-push) factors and **Inflation Expectations**.

1. Demand-Side Causes (Demand-Pull Inflation)

Demand-pull inflation occurs when aggregate demand exceeds aggregate supply. It is often described as "too much money chasing too few goods."

1.1 Increase in Money Supply

When the central bank increases the money supply, consumers and firms have more money to spend, which boosts demand and pushes prices up.

Example: Government stimulus programs financed by printing money.

1.2 Increase in Government Spending

Government expenditure on infrastructure, wages, or subsidies increases total demand in the economy.

Example: Public investment in roads and hospitals.

1.3 Increase in Consumer Spending

High consumer confidence or access to credit can lead to increased spending on goods and services.

Example: Holiday season spending.

1.4 Increase in Business Investment

Firms investing in capital goods (machines, equipment) increases demand for such products.

1.5 Increase in Exports

Higher demand for domestic goods from abroad can reduce local supply and drive up prices.

Example: A surge in agricultural exports.

2. Supply-Side Causes (Cost-Push Inflation)

Cost-push inflation occurs when the cost of production increases, leading businesses to raise prices to maintain profit margins.

2.1 Increase in Wages

When workers negotiate higher wages without productivity gains, firms may pass these costs to consumers.

2.2 Increase in Raw Material Prices

Increased prices for inputs like oil or metals raise production costs.

Example: Global oil price surge affecting fuel and transport costs.

2.3 Natural Disasters

Disasters can reduce supply by damaging infrastructure, factories, or crops.

Example: Floods destroying food crops.

2.4 Currency Devaluation

A fall in the value of a country's currency increases the cost of imported goods and services.

Example: A weaker birr making foreign machinery more expensive.

3. Inflation Expectations

Though not always listed separately, inflation expectations are crucial. If workers and firms expect future inflation, they may act in ways that cause it:

- Workers demand higher wages.
- Firms increase prices in anticipation of higher costs.

This creates a **self-fulfilling cycle** of inflation.

Types of Inflation

1. Inflation Based on Rate

- **Creeping Inflation:** This is a mild and gradual increase in the price level, usually less than 3% annually. It is generally considered manageable and sometimes beneficial for economic growth.
- **Walking Inflation:** Inflation rate between 3% and 10% per year. This indicates a growing inflationary pressure that requires monitoring as it can affect economic decisions.
- **Galloping Inflation:** Rapid inflation, often running into double or triple digits per year (e.g., 10% to 100%). It causes uncertainty in the economy, discourages savings, and distorts investment.
- **Hyperinflation:** Extremely high inflation, often exceeding 50% per month. It usually results from excessive money printing, loss of confidence in currency, and severe economic crises. Hyperinflation causes currency devaluation and collapse of the financial system.

2. Inflation Based on Causes

- **Demand-Pull Inflation:** Occurs when aggregate demand exceeds aggregate supply in the economy. Causes include increased consumer spending, government expenditure, or investment, leading to upward pressure on prices.
- **Cost-Push Inflation:** Caused by an increase in the cost of production factors such as wages, raw materials, or energy. Producers pass these higher costs to consumers through price increases.
- **Built-in Inflation:** Also called wage-price inflation, it arises from adaptive expectations. Workers demand higher wages to keep up with past inflation, and businesses raise prices to cover higher wage costs, creating a wage-price spiral.

3. Inflation Based on Coverage

- **General Inflation:** Inflation that affects the general price level across most sectors of the economy.
- **Sectoral Inflation:** Inflation limited to specific sectors or commodities. For example, food inflation or housing inflation may rise independently due to sector-specific factors.

4. Inflation Based on Occurrence

- **Demand Inflation:** Inflation triggered primarily by excess demand relative to supply in the economy.
- **Supply Inflation:** Caused by disruptions in supply chains, shortage of raw materials, or increase in production costs which reduce supply or increase prices.

5. Inflation Based on Government Reaction

- **Open Inflation:** Occurs when prices rise openly in the market without government intervention. It usually happens in a free market without price controls.

- **Suppressed Inflation:** Happens when the government imposes price controls, rationing, or subsidies to prevent prices from rising. Though prices may appear stable, inflationary pressures accumulate underground and may resurface later.

1.3 Annuities

Definition: An annuity is a series of equal payments made at regular intervals for a set period.

Payments of any type are considered as **annuities** if all of the following conditions are present:

- The periodic payments are equal in amount.
- The time between payments is constant (such as a year, half a year, a quarter, a month, etc.).
- The interest rate per period remains constant.
- The interest is compounded at the end of every period.

These conditions are essential to apply standard annuity formulas in financial mathematics for calculating present value, future value, and payment schedules under consistent interest compounding.

Annuities involve several important terminologies:

1. **Payment interval or period:** The time between successive payments of an annuity.
2. **Term of annuity:** The period or time interval between the beginning of the first payment period and the end of the last payment period.
3. **Conversion or interest period:** The interval between consecutive conversions of interest.
4. **Periodic payment (rent):** The amount paid at the end or the beginning of each payment period.
5. **Simple annuity:** An annuity in which the payment period and the conversion period coincide with each other.

These terminologies will be used throughout annuity discussions when calculating present value, future value, and payment structures in financial mathematics.

Types of annuity:

- Ordinary Annuity
- Annuity Due

Ordinary Annuity: An annuity where payments are made at the **end of each period**.

Annuity Due: An annuity where payments are made at the **beginning of each period**.

For both types, the Future Value (FV) and Present Value (PV) are computed using:

Future Value of Ordinary Annuity:

$$FV = PMT \times \frac{((1+r)^n - 1)}{r}$$

Future Value of Annuity Due:

$$FV_{AD} = FV \times (1+r)$$

or directly:

$$FV_{AD} = PMT \times \frac{((1+r)^n - 1)}{r} \times (1+r)$$

where:

- PMT: payment per period
- r: interest rate per period
- n: total number of payments

Examples of Ordinary Annuity:

Example 1

You save $1,000 yearly at 7% annual interest, paid at the end of each year, for 3 years. Find the future value.

Solution:

$$FV = 1000 \times \frac{(1.07^3 - 1)}{0.07} = 1000 \times \frac{0.225043}{0.07} = 1000 \times 3.215 = 3215$$

FV = $3,215

Example 2

If you deposit $250 per month at the end of each month in an account paying 0.5% monthly for 2 years, find the future value.

Solution:

$$r = 0.005, \quad n = 24$$

$$FV = 250 \times \frac{(1.005^{24} - 1)}{0.005} = 250 \times \frac{0.127159}{0.005} = 250 \times 25.4318 = 6357.95$$

FV = $6,357.95

Example 3

You invest \$400 annually at 4% annually for 6 years, with payments at the end of each year. Find the future value.

Solution:

$$FV = 400 \times \frac{(1.04^6 - 1)}{0.04} = 400 \times \frac{0.265319}{0.04} = 400 \times 6.63297 = 2653.19$$

FV = \$2,653.19

Examples of Annuity Due

Example 1

You deposit \$500 at the beginning of each year into an account paying 5% annual interest for 4 years. Find the future value.

Solution: First, compute FV as an ordinary annuity:

$$FV = 500 \times \frac{(1.05^4 - 1)}{0.05} = 2155.06$$

Then:

$$FV_{AD} = 2155.06 \times 1.05 = 2262.81$$

$FV_{AD} = \$2,262.81$

Example 2

You invest \$300 per year at the beginning of each year in an account earning 6% annually for 5 years. Find the future value.

Solution: Ordinary FV:

$$FV = 1691.13$$

$$FV_{AD} = 1691.13 \times 1.06 = 1792.60$$

$FV_{AD} = \$1,792.60$

Example 3

You save \$1,000 yearly at 7% annual interest, payments at the beginning of each year for 3 years. Find the future value.

Solution: Ordinary FV:

$$FV = 3215$$

$$FV_{AD} = 3215 \times 1.07 = 3440.05$$

$FV_{AD} = \$3,440.05$

Example 4

If you deposit \$250 per month at the beginning of each month at 0.5% monthly for 2 years, find the future value.

Solution: Ordinary FV:

$$FV = 6357.95$$

$$FV_{AD} = 6357.95 \times 1.005 = 6389.73$$

$FV_{AD} = \$6,389.73$

Example 5

You invest \$400 annually at 4% annually for 6 years with payments at the beginning of each year. Find the future value.

Solution: Ordinary FV:

$$FV = 2653.19$$

$$FV_{AD} = 2653.19 \times 1.04 = 2759.32$$

$FV_{AD} = \$2,759.32$

1.4 Bonds

Definition: A bond is a fixed-income security representing a loan from an investor to a borrower, paying periodic interest and repaying the principal at maturity.

A bond has:

- **Face Value (Par Value):** The amount the issuer agrees to pay back at maturity, typically \$1,000.
- **Coupon Rate:** The annual interest rate paid on the bond's face value.
- **Maturity:** The date when the face value is repaid.
- **Yield:** The rate of return based on the bonds current price.

Example: A bond has:

- Face value: \$1,000
- Coupon rate: 6%
- Maturity: 5 years

Solution: You receive \$60 annually for 5 years and \$1,000 at maturity.

Types of Bonds

- **1. Government Bonds:** Issued by governments (e.g., Treasury Bonds).
  Example: U.S. Treasury Bond (T-Bond), Ethiopian Government Development Bond.
- **2. Corporate Bonds:** Issued by corporations to raise capital.
  Example: Apple Inc. corporate bond, Ethiopian Airlines corporate bond.
- **3. Zero-Coupon Bonds:** Sold at a discount and pay no periodic interest; the investor receives the face value at maturity.
  Example: U.S. Treasury STRIPS, Zero-coupon bonds issued by government or large corporations.
- **4. Callable Bonds:** The issuer can repay before maturity.
  Example: AT&T Callable Bond the company can redeem it before the maturity date.
- **5. Convertible Bonds:** Can be converted into a predetermined number of shares of the issuing company.
  Example: Tesla convertible bonds can be converted into Tesla shares.

Example 1: Calculating Annual Coupon Payment

A bond has a face value of \$1,000 and a coupon rate of 6%. What is the annual coupon payment?

Solution:

$$\text{Coupon Payment} = \text{Face Value} \times \text{Coupon Rate} = 1000 \times 0.06 = 60$$

The annual coupon payment is \$60.

Example 2: Total Interest Earned Over Holding Period

You buy a bond with a face value of \$1,000, a 5% annual coupon rate, and 4 years to maturity. How much total interest will you receive if you hold the bond until maturity?

Solution: Annual interest:

$$1000 \times 0.05 = 50$$

Total interest over 4 years:

$$50 \times 4 = 200$$

Total interest received will be \$200.

Example 3: Zero-Coupon Bond

A zero-coupon bond has a face value of \$1,000 and matures in 5 years. If it is sold for \$800, find the implied annual yield.

Solution: We use:

$$FV = PV \times (1+r)^t$$

where

- FV = Face value of the bond = \$1,000
- PV = Present value or purchase price of the bond = \$800
- r = Implied annual yield (interest rate) [to be found]
- t = Time to maturity in years = 5

$$1000 = 800 \times (1+r)^5$$
$$(1+r)^5 = \frac{1000}{800} = 1.25$$

Take the 5th root:

$$1 + r = 1.25^{1/5} = 1.0456$$
$$r = 0.0456 = 4.56\%$$

The implied annual yield is 4.56%.

Example 4: Price of a Bond (Present Value)

A bond pays a \$40 annual coupon, has 3 years to maturity, and the market yield is 5%. Find the price of the bond (assuming face value is \$1,000).

Solution: Bond price:

$$P = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{FV}{(1+r)^n}$$

Where:

- P = Price of the bond (Present Value) [to be found]
- C = Annual coupon payment = \$40
- FV = Face value of the bond = \$1,000
- r = Market interest rate (yield) = 5% = 0.05
- n = Number of years to maturity = 3

So:

$$C = 40, \quad r = 0.05, \quad n = 3, \quad FV = 1000$$

Calculation:

$$P = \frac{40}{1.05} + \frac{40}{1.1025} + \frac{40}{1.157625} + \frac{1000}{1.157625}$$
$$P = 38.10 + 36.30 + 34.54 + 863.48 = 972.42$$

The price of the bond is \$972.42.

Example 5: Yield to Maturity (YTM)

A bond with a face value of $1,000 pays an annual coupon of $50 and has 2 years to maturity. If its current price is $950, find the approximate yield to maturity.

Solution: Approximate YTM formula:

$$YTM \approx \frac{C + \frac{FV-P}{n}}{\frac{FV+P}{2}}$$

Where:

- YTM = Approximate yield to maturity [to be found]
- C = Annual coupon payment = $50
- FV = Face value of the bond = $1,000
- P = Current price of the bond = $950
- n = Number of years to maturity = 2

Given:

$$C = 50, \quad FV = 1000, \quad P = 950, \quad n = 2$$

Calculation:

$$YTM \approx \frac{50 + \frac{1000-950}{2}}{\frac{1000+950}{2}} = \frac{50+25}{975} = \frac{75}{975} = 0.0769 = 7.69\%$$

The approximate yield to maturity is 7.69%.

1.5 Internal Rate of Return (IRR)

Definition: IRR is the discount rate that makes the Net Present Value (NPV) of all cash flows of a project or investment equal to zero.

$$0 = \sum_{t=0}^{n} \frac{C_t}{(1+r)^t}$$

Where:

- C_t = Cash flow at time period t
- r = Internal Rate of Return (IRR)
- t = Time period (year)
- n = Total number of periods
- C_0 = is usually a negative number representing the initial investment

The IRR helps determine whether to accept or reject an investment:

- If $IRR >$ required rate of return, accept the investment.
- If $IRR <$ required rate of return, reject the investment.

Example 1

An investment of \$1,000 returns \$1,100 after one year. Find the IRR.

Solution:

We need to solve:

$$0 = -1000 + \frac{1100}{(1+r)}$$

$$1000 = \frac{1100}{(1+r)}$$

$$1 + r = \frac{1100}{1000} = 1.1$$

$$r = 0.1 = 10\%$$

The IRR is 10%.

Example 2

An investment of \$2,000 yields \$1,200 at the end of year 1 and \$1,000 at the end of year 2. Find the IRR.

Solution:

$$0 = -2000 + \frac{1200}{(1+r)^1} + \frac{1000}{(1+r)^2}$$

Try $r = 0.1 = 10\%$:

$$NPV = -2000 + \frac{1200}{1.1} + \frac{1000}{1.21} = -2000 + 1090.91 + 826.45 = -82.64$$

Try $r = 0.08 = 8\%$:

$$NPV = -2000 + \frac{1200}{1.08} + \frac{1000}{1.1664} = -2000 + 1111.11 + 857.34 = -31.55$$

Try $r = 0.06 = 6\%$:

$$NPV = -2000 + \frac{1200}{1.06} + \frac{1000}{1.1236} = -2000 + 1132.07 + 890.84 = 22.91$$

By interpolation:

$$IRR \approx 8\% + \frac{31.55}{31.55 + 22.91} \times (6\% - 8\%) = 8\% - \frac{31.55}{54.46} \times 2\% = 8\% - 1.16\% = 6.84\%$$

The IRR is approximately 6.8%.

Example 3

You invest \$5,000 and receive \$2,000 per year for 3 years. Find the IRR.

Solution:

$$0 = -5000 + \frac{2000}{(1+r)^1} + \frac{2000}{(1+r)^2} + \frac{2000}{(1+r)^3}$$

Try $r = 0.10 = 10\%$:

$$NPV = -5000 + \frac{2000}{1.1} + \frac{2000}{1.21} + \frac{2000}{1.331} = -5000 + 1818.18 + 1652.89 + 1503.37 = -25.56$$

Try $r = 0.09 = 9\%$:

$$NPV = -5000 + \frac{2000}{1.09} + \frac{2000}{1.1881} + \frac{2000}{1.2950} = -5000 + 1834.86 + 1683.46 + 1544.88 = 63.20$$

By interpolation:

$$IRR \approx 9\% + \frac{63.20}{63.20 + 25.56} \times (10\% - 9\%) = 9\% + \frac{63.20}{88.76} \times 1\% = 9\% + 0.71\% = 9.71\%$$

The IRR is approximately 9.7%.

Example 4

An investment of \$4,000 returns \$2,000 at the end of year 1 and \$2,500 at the end of year 2. Find the IRR.

Solution:

$$0 = -4000 + \frac{2000}{(1+r)} + \frac{2500}{(1+r)^2}$$

Try $r = 0.15 = 15\%$:

$$NPV = -4000 + \frac{2000}{1.15} + \frac{2500}{1.3225} = -4000 + 1739.13 + 1890.73 = -370.14$$

Try $r = 0.10 = 10\%$:

$$NPV = -4000 + \frac{2000}{1.1} + \frac{2500}{1.21} = -4000 + 1818.18 + 2066.12 = -115.70$$

Try $r = 0.05 = 5\%$:

$$NPV = -4000 + \frac{2000}{1.05} + \frac{2500}{1.1025} = -4000 + 1904.76 + 2268.24 = 173$$

By interpolation:

$$IRR \approx 5\% + \frac{173}{173 + 115.7} \times (10\% - 5\%) = 5\% + \frac{173}{288.7} \times 5\% = 5\% + 3\% = 8\%$$

The IRR is approximately 8%.

Example 5

An investment of \$3,000 yields \$1,000 at the end of each year for 4 years. Find the IRR.

Solution:

$$0 = -3000 + \frac{1000}{(1+r)} + \frac{1000}{(1+r)^2} + \frac{1000}{(1+r)^3} + \frac{1000}{(1+r)^4}$$

Try $r = 0.15 = 15\%$:

$$NPV = -3000 + \frac{1000}{1.15} + \frac{1000}{1.3225} + \frac{1000}{1.5209} + \frac{1000}{1.7490} = -3000 + 869.57 + 756.14 + 657.51 + 571.94 = -144.84$$

Try $r = 0.12 = 12\%$:

$$NPV = -3000 + \frac{1000}{1.12} + \frac{1000}{1.2544} + \frac{1000}{1.4049} + \frac{1000}{1.5735} = -3000 + 892.86 + 797.19 + 711.98 + 635.66 = 37.69$$

By interpolation:

$$IRR \approx 12\% + \frac{37.69}{37.69 + 144.84} \times (15\% - 12\%) = 12\% + \frac{37.69}{182.53} \times 3\% = 12\% + 0.62\% = 12.62\%$$

The IRR is approximately 12.6%.

Example 6

An investment costs \$1,000, returning \$600 per year for 2 years. Find the IRR.

Solution: Solve:

$$0 = -1000 + \frac{600}{(1+r)^1} + \frac{600}{(1+r)^2}$$

By trial:

- At $r = 0.13$:

$$= -1000 + \frac{600}{1.13} + \frac{600}{1.2769} = -1000 + 531.86 + 470.07 = 1.93$$

Thus, IRR is approximately 13%.

Remark

Definition Linear Interpolation

Given two known points (x_0, y_0) and (x_1, y_1), the linear interpolation formula is:

$$y = y_0 + \frac{(x - x_0)(y_1 - y_0)}{x_1 - x_0}$$

This formula estimates the value y at a point x between x_0 and x_1, assuming the change between the two points is linear.

Example

Suppose we are given two points:

$$(x_0, y_0) = (2, 4), \quad (x_1, y_1) = (6, 8)$$

Estimate the value of y when $x = 4$.

Using the formula:

$$y = 4 + \frac{(4-2)(8-4)}{6-2} = 4 + \frac{2 \cdot 4}{4} = 4 + 2 = 6$$

Therefore, the interpolated value at $x = 4$ is $y = 6$.